Code: DMS-06 *Paper ID: 25661*

GNIOT Institute of Management Studies – PGDM (Batch 2023-25)

(TRIMESTER – V) END TERM EXAMINATION, JANUARY -2025

SUPPLY CHAIN ANALYTICS

[Time: 2 Hours] ***[Maximum Marks: 60]***

Following are the course outcomes of the subject: - SUPPLY CHAIN ANALYTICS

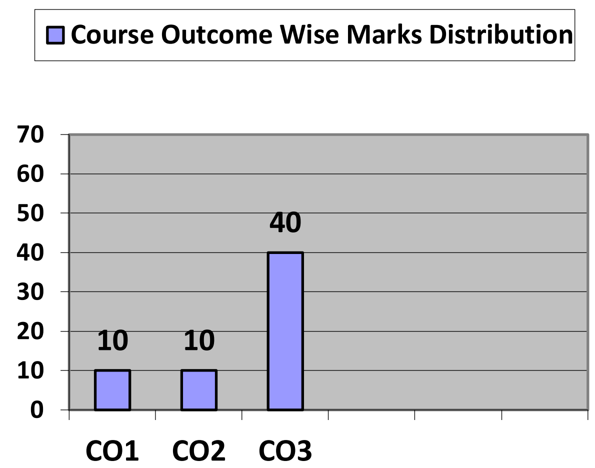
CO	Course Outcome (CO)
CO1	To understand the different supply chain optimization techniques
CO2	To understand the concept of Modelling and it use in Supply chain.
CO3	To understand the use of Linear Programming technique to support decision-making processes in supply chain management.

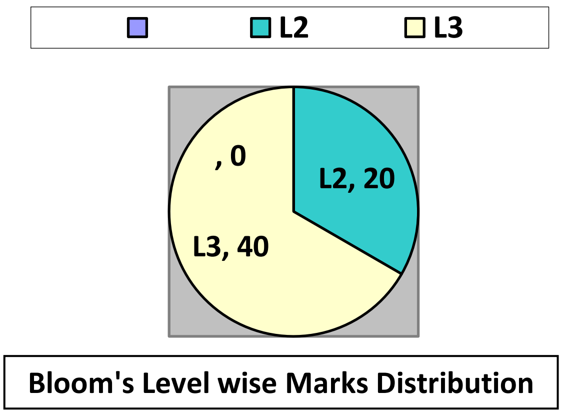
(Note- Faculty can add the more CO's as per their subject.)

SECTION-A

Attempt all parts of the following: (Short Answer Type Questions, do not exceed 150- 200 words)

Q	Marks	COURSE OUTCOME	BLOOMS LEVEL
(1)	**10**	CO - 1	BL – 2
(2)	**10**	CO – 2	BL – 2
(3)	**10**	CO – 3	BL – 3
(4)	**10**	CO – 3	BL- 3
(5)	**10**	CO – 3	BL- 3
(6)	**10**	CO - 3	BL- 3





Moderated By **IQAC** **Dean - Examinations &PGP**

Printed Pages: 1

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[Time: 2 Hours] ***[Maximum Marks: 60]***

INSTRUCTIONS:

(i) All questions are mandatory

(ii) All questions carry equal marks i.e. 10 marks each.

(iii) Kindly write down mathematical formulation & network diagram (wherever applicable) in the answer sheet.

(iv) After solving the LP problem, write the value of Decision Variable and Objective Function in the answer sheet.

(v) Kindly create separate sheet for each question

(vi) Save your excel file as "Student ID_Name" and upload the same in the link shared at the end of the examination.

Q1: GIMS Company produces high-quality Sweaters and Jackets for Women. Each pair of Sweaters requires 3.6 hours of cutting time and 2.1 hours of sewing time, uses 21 yards of material and provides a profit contribution of $210. Each Jacket needs 2.6 hours of cutting time and 2.1 hours of sewing time, uses 12 yards of material and provides a profit contribution of $112. For the coming month, 600 hours of cutting time, 630 hours of sewing time and 2600 yards of fabric are available. The marketing department specifies a minimum production of 95 sweaters and 115 Jackets. Assume that GIMS Company is interested in maximizing the total profit contribution. Use the following steps to find the best decision through LP.

Q) Formulate a linear programming model (mathematical Function) for this problem. Determine how many trousers and shirts should be produced to maximize the total profit, and the total profit.

Q2: Smart Tech Corporation manufactures three types of circuit boards: Alpha, Beta, and Gamma. The company recently received a large order for 5,000 Alpha boards, 3,000 Beta boards, and 2,000 Gamma boards. Each board requires a certain number of design hours and testing hours, summarized in the table below:**

	Alpha	Beta	Gamma
Number Ordered	5,000	3,000	2,000
Design Hours/Unit	1	2	3
Testing Hours/Unit	2	1.5	2

Unfortunately, Smart Tech has limited resources. The company only has 10,000 design hours and 9,000 testing hours available for this order. Smart Tech can manufacture the boards in-house or subcontract part of the order to a competitor. The unit costs are summarized below:

	Alpha	Beta	Gamma
Cost to Make	$25	$40	$60
Cost to Buy	$35	$50	$75

Smart Tech wants to determine how many of each board type to manufacture in-house and how many to subcontract in order to minimize the total cost of fulfilling the order.

Q) Formulate a Linear Programming model (mathematical Function) for this scenario. Define: Decision variables, Objective function (minimize cost), Constraints (demand, resource limitations, non-negativity). Also Solve the problem using any optimization method or software.

Q3: Global Logistics Network (GLN) specializes in planning optimal delivery routes for its clients. GLN's system identifies the shortest or most efficient routes for goods to be transported between warehouses and retail outlets across various locations. GLN has recently been tasked with finding the optimal route for a shipment traveling from Warehouse A to Retail Outlet F. The network of cities and routes between Warehouse A and Retail Outlet F is shown in the graph below. Each route is labeled with the travel time (in hours) and the reliability score (higher scores indicate better route conditions and safety).

The nodes represent cities, and the arcs represent the routes between cities. The goal is to either:

1. Minimize travel time for the shipment (shortest path problem), or
2. Maximize the reliability score (most reliable path problem).

Route	Travel Time (hours)	Reliability Score
A → B	Last Digit of your ID	8
A → C	4.5	6
B → D	2.5	9
C → D	3	7
B → E	3.5	5
D → E	First Digit of your ID	8
D → F	3	7
E → F	1.5	9

At **Node A (Warehouse A): Supply = +1** (1 unit of flow is being supplied).
At **Node F (Retail Outlet F): Demand = -1** (1 unit of flow is required here).

Q) Construct a Shortest path Network Diagram. Formulate the problem as a shortest path problem to minimize travel time from Warehouse A to Retail Outlet F. Solve the problem using LP.

Q4: National Freight Logistics manages a supply chain for distributing goods between various ports (supply points) and major distribution hubs (demand points). Each port has a certain capacity of goods available, and each hub has a specific demand for goods. The transportation cost per unit from each port to each hub is provided below.

Hub/Port	Jaipur	Lucknow	Nagpur	Indore	Surat	Port Supply
Vizag	6	8	5	4	7	1st 5 Digit of your student ID
Cochin	7	6	4	3	8	12,500
Paradip	5	9	6	7	9	Last 5 Digit of your student ID
Tuticorin	8	7	5	6	6	11,000
Demand	Last 4 Digit of your Student ID	7500	6000	1st 4 Digit of your student ID	4500	

Q) Formulate the problem as a Linear Programming (LP) model. Construct a visual Network Design. Solve the LP model using an optimization tool

Q5: The distribution system of Lloyd Electronic Company consist of 3 warehouses & 5 customers.
Using the given information below:

	Warehouse			
Plant	**1**	**2**	**3**	**Capacity**
A	5	9	5	650
B	7	3	6	800
C	4	8	8	960

	Customer				
Warehouse	**1**	**2**	**3**	**4**	**5**
1	3	5	7	9	12
2	5	6	11	5	14
3	2	8	5	4	5
Demand	**350**	**460**	**380**	**530**	**400**

Q) Develop a network representation of this problem. Formulate a LP model of the problem. Solve the LP to determine the optimal shipping plan.

Q6: Suppose that the Tech Gadgets Store sells a popular wireless headphone product with a constant annual demand of 4,800 units. Each unit costs the store $50 to purchase from its supplier. The store incurs ordering costs of $40 per order and holding costs are 20% of the unit cost per year. The store operates 300 working days per year, and the supplier requires a lead time of 7 days to fulfil an order.

Q) Identify the following aspects of the inventory policy:

a. Economic Order Quantity (EOQ)
b. Reorder Point (ROP)
c. Cycle Time (time between orders in days)
d. Total Annual Inventory Cost